CMPT 276 Phase 4 Report

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1. The Game

Our game is a 2D grid-based dungeon game where the player moves through a maze, collects keys, opens chests, and avoids enemies and traps. The main goal is to collect all keys and reach the exit.

The original inspiration for Phase 1 was “dungeon crawler rougelike”. The player would navigate through a map filled with spike traps, goblins, and ogres, while collecting keys and optional rewards such as chests. Goblins were designed to chase the player, while ogres would follow fixed patrol paths. The game also uses a tick-based system where the game state advances on each tick and enemies move on each tick.

Overall, our final game stayed very faithful to this original plan. The core gameplay loop of collecting keys, avoiding enemies, and reaching the exit was fully implemented as intended. The enemy behaviors were also implemented as planned, with goblins chasing the player and ogres following predictable movement paths.

However, there were some changes and improvements made during development. One improvement was the addition of popup messages. These were not part of the original design, but were added later to improve the user experience. For example, when the player collects a key or collects a reward, a popup appears to inform them. Similarly, a game over popup appears when the player loses. These additions made the game easier to understand and more interactive.

Another improvement was the addition of a pause feature. This was also not part of the original design. During gameplay, the player can press P to pause the game. This gives the player more time to think or take a short break.

We also improved how rewards work. Chests still appear after collecting keys as planned, but their behavior was refined during implementation to better fit the pacing of the game.

One of the most important lessons we learned from this project is that initial designs will almost always need to change. Even though our Phase 1 plan was solid, we had to adjust parts of it once we started implementing and testing the game. For example, adding popups, adding a pause feature, and refining reward behavior were decisions made after seeing how the game actually felt during gameplay.

Another key lesson was the importance of communication and consistency when working in a group. While we were able to divide the work into separate parts, integrating everything together was more difficult than expected. This taught us that having consistent design decisions and clear communication is just as important as writing the code itself.

Overall, we are satisfied with how the final game turned out. It matches our original idea closely while also including improvements that made the gameplay clearer and more enjoyable.

2. Tutorial

This section provides a short tutorial on how to play the game and highlights the main features.

The player controls a character that moves around the grid-based dungeon. The objective is to collect all keys and then reach the exit.

- Move around the map using keyboard controls (WASD or arrow keys)
- Press P to pause the game
- Collect all keys to unlock the exit
- Avoid enemies such as goblins and ogres
- Avoid spike traps, which reduce your score
- Collect chests for bonus points when they appear

Video Demo

https://youtu.be/Cf1U1OJiNrQ

Gameplay Features

- Pause: The player can press P during gameplay to pause the game. Pressing Space resumes the game

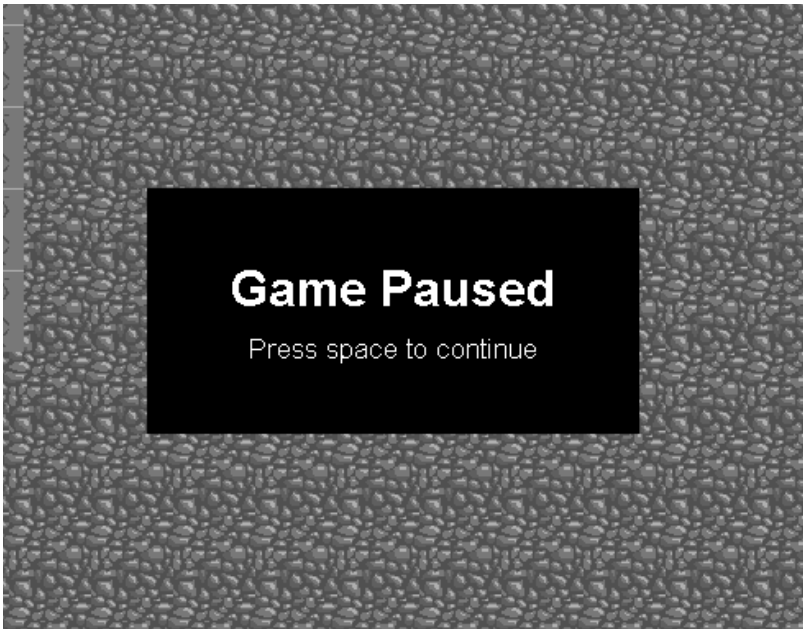

Figure 1: Pause Screen

- Key collection: Keys are required to win the game. The player must collect all keys before they can pass through the exit and win

Figure 2: Player Collecting a Key

- Goblins: Goblins chase the player. A collision with a goblin results in a game over

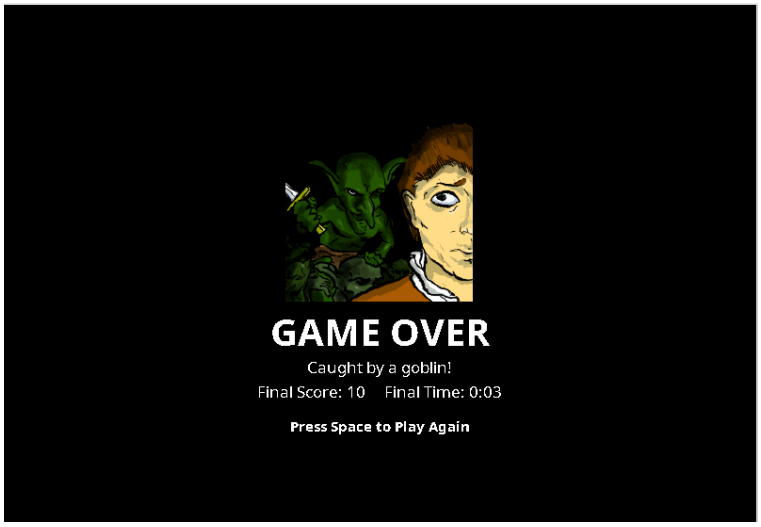

Figure 3: Game Over via Collision with Goblin Enemy

- Ogres: Ogres follow fixed patrol paths. A collision with an ogre reduces the player's score. If the player's score falls below zero, the player loses the game

Figure 4: Collision with Ogre Enemy

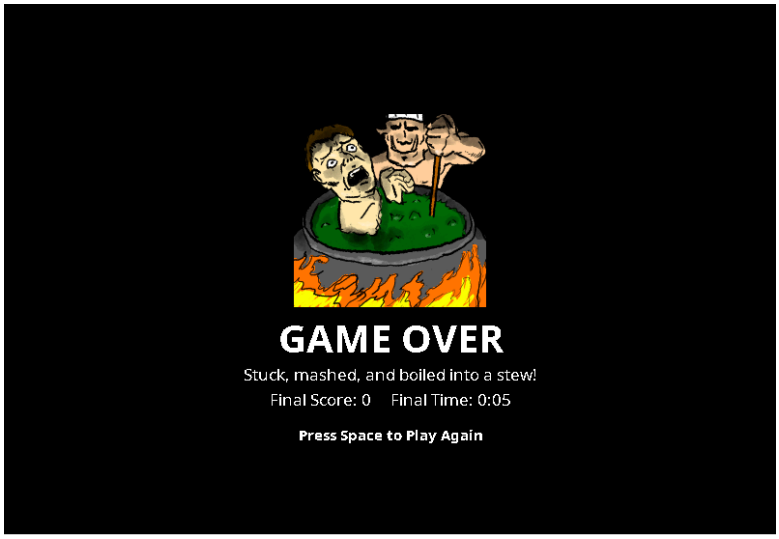

Figure 5: Game Over via Collision with Ogre Enemy at Low Score

- Spike Traps: Decrement the player's score. If the player's score falls below zero, they'll die to the spike trap

Figure 6: Game Over via Collision with Spike Trap with Low Score

- Rewards: Bonus rewards (treasure chests) spawn in intervals for a limited time. Collecting it increases the player's score

Figure 7: Player Collecting a Bonus Reward

- Victory: After collecting all keys the player can pass through the exit on the map

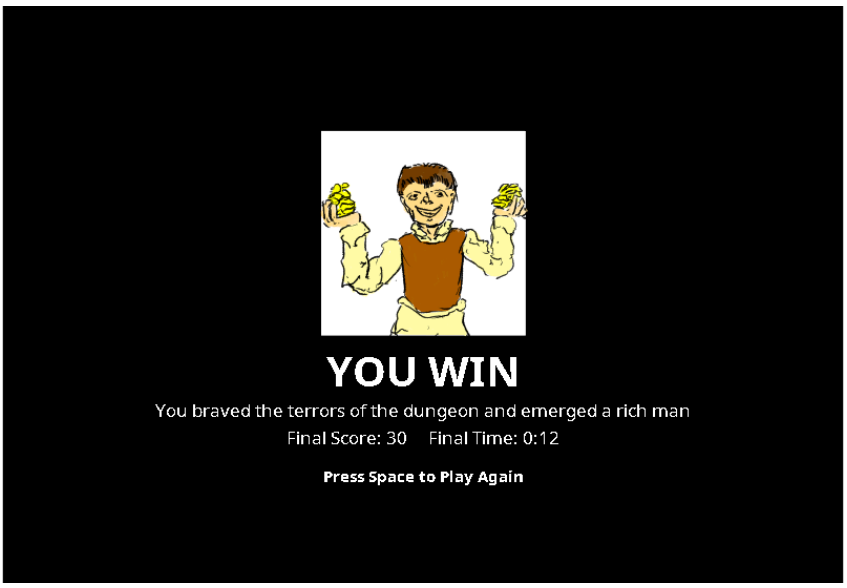

Figure 8: Victory Screen

Generated Artifacts

The generated artifacts have also been added to the repository.
The executable game JAR file is located at: **Artifacts/spring2026team7-1.0.jar**
The generated Javadocs can be accessed at: **Artifacts/apidocs/index.html**